Lab 1 Report

System Identification

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1 Introduction

Background and Context

Passenger comfort and safety are kept as top priorities in aerospace engineering. Aircraft passengers often experience insecurity and discomfort during landings and takeoffs due to shocks and vibrations. To fix these issues, shock absorbers are added to the landing gear. These components are the ones that play a huge role in dissipating energy, reducing vibrations, and resulting in a smoother ride, but a well-designed suspension system also protects the structural integrity of the aircraft, reducing maintenance demand.

Objective of the Experiment

This lab aims to find the transfer function of an aircraft landing gear suspension system using time-domain (step response) and frequency-domain (sinusoidal input) tests. In analysing this system's response, the parameter required to model a suspension system can be found. Once this is done, it will be possible to validate it against simulated landing and take-off conditions. These parameters are the damping ratio, natural frequency, and overall system gain. This approach provides valuable insight into how the suspension behaves under different loading scenarios, allowing engineers to evaluate performance, improve stability, and ensure passenger comfort without the need for costly or risky full-scale testing

Scope of the Experiment

The primary objective of this experiment is to use a dedicated simulation software to examine the dynamic behavior of the aircraft suspension system. This tool provides an interactive platform for modelling and studying different aspects of the suspension, focusing particularly on displacement, applied forces, and overall system response under a range of conditions.

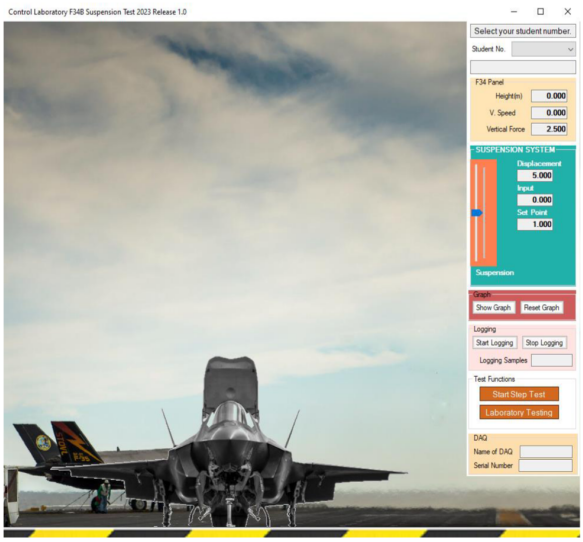

Figure 1: Control Laboratory Suspension Simulator

Figure 1 shows the graphical user interface of the simulation program used in this experiment. The interesting thing with this interface is that it allows for adjustments to

parameters such as set point, input force, and displacement while it shows a real-time visualization of the system's behavior simultaneously, which allows precise fine-tuning of these variables and collections and data necessary for a thorough understanding of the suspension system's performance.

2 System Modelling

2.1 Methodology

1. A simulation environment was set up using the control Laboratory34B Suspension Simulator.
2. A unit step input was applied to the system via the offset input feature, and the system's behaviour(response) was recorded in a spreadsheet to analyse transient and steady-state behaviour.
3. Multiple sinusoidal inputs with different frequencies were applied to the system to study the response and construct a Bode plot.
4. Data from both of the tests were used to calculate the system gain, damping ratio, natural frequency, spring constant, and damping constant.
5. Finally, the found transfer function was simulated in MATLAB and compared to the experimental results for validation and refinement of the model.

2.2 System Model

2.2.1 A differential equation describing the suspension system

The equation describing this can be seen as a second–order equation which is:

$$\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2\, x(t) = A\,\omega_n^2\, F(t). \tag{1}$$

where

2.2.2 A transfer function derived from the differential equation

Taking the Laplace transform of the differential equation

$$(s^2 + 2\zeta\omega_n s + \omega_n^2)X(s) = A\,\omega_n^2\, F(s).$$

Therefore, the transfer function from input $F(s)$ to output $X(s)$ is

$$G(s) = \frac{X(s)}{F(s)} \;=\; \frac{A\,\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \tag{2}$$

.

3 System Identification

3.1 Step Response Test Stage1

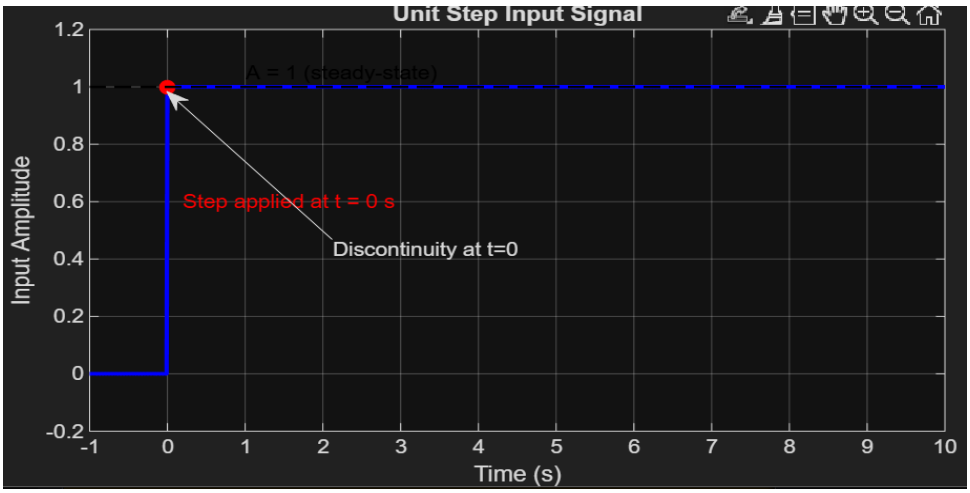

Figure 2: Step input to the suspension system to achieve a step response

To examine the dynamic behavior of the aircraft suspension system, a step input was applied as mentioned in the methodology, and the system to respond with and the resulting response was recorded over a time interval [0s-8s]. Figure 3 illustrates the step response which came out as resulted from the unit step input shown in Figure 1. The step input represents a sudden change in the system's input from 0 to the offset, introduced through the offset input feature in the suspension simulator.

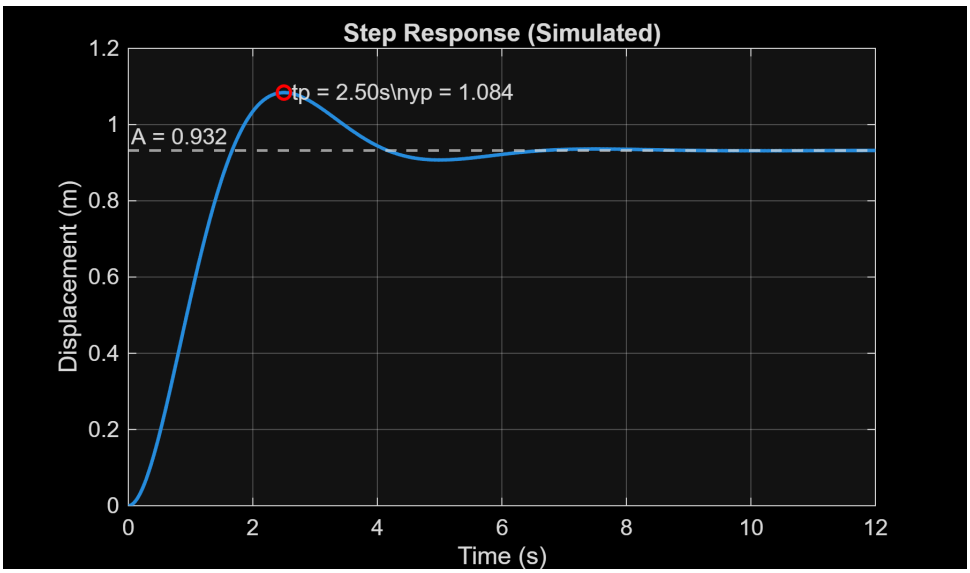

Figure 3: Step response of suspension system: showing overshoot, settling time, and steady-state value.

Now looking into this step response, shown in Figure 3, we can find the transfer function of the system by analyzing the dynamic characteristics of the system.

Now, from the response shown in Figure 3, we can answer the following question regarding the system:

1. What is the gain of the system? Answer: Since the gain of a system is the ratio of the steady-state output to the input, it reflects how much the system amplifies or attenuates the input signal. Through observation response in Figure 3, we see that at steady state, so $A = \lim_{t\to\infty} y(t)$ resulting in the gain equaling 0,983; therefore, A = 0.932
2. Does the system have real or complex poles? Answer: From Figure 3 we see that the step response oscillates, telling us that the system is underdamped, so the poles are complex.
3. Where do you estimate the dominant pole or complex pole pair to be? Answer: The dominant pole pair is expected to be on the left of the S-plane and close to the imaginary axis. Its exact position is dependent on the system's natural frequency and damping ratio, which dictates how oscillatory and quickly decaying a response is.
4. From (3), what would you expect the corner frequency of this system to be? Answer: From the data presented in Table 1, it is possible to estimate the corner frequency of the underdamped second-order system depicted in Figure 3.

Table 1: Samples from the Step Test data

Time [s]	**Input**	**Output [m]**
...	...	...
2.5	1	1.084
...	...	...
7.4	1	0,936
...	...	...

From Table 1, the step response peaks at $t_1 = 2.5$ s and $t_2 = 7.4$ s, the period is $T = 7.7s - 2.5s = 5.2$ s, hence

$$f = \frac{1}{T} \approx 0.1923\ Hz, \qquad \omega_c = 2\pi f \approx 1.208\ rad/s.$$

Calculation of the System Identification Values;

$$t_p = \frac{\pi}{\omega_d}, \quad \omega_d = \frac{\pi}{2.5} \approx 1.257\, rad/s$$

$$y_p = A\left(1 + e^{-\zeta\pi\sqrt{1-\zeta^2}}\right), \quad \frac{1.084}{0.932} - 1 = 0.163 = e^{-\zeta\pi\sqrt{1-\zeta^2}}$$

$$\frac{\zeta\pi}{\sqrt{1-\zeta^2}} \approx 1.813 \quad \Rightarrow \quad \zeta \approx 0.499$$

$$\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}} = \frac{1.257}{\sqrt{1-0.499^2}} \approx 1.447\, rad/s$$

$$G(s) = \frac{A\,\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{0.932 \times (1.447)^2}{s^2 + 2(0.499)(1.447)s + (1.447)^2}$$

$$G(s) = \frac{1.95}{s^2 + 1.44s + 2.10}$$

Now, from what is done, this system can be modelled as a spring-mass-damper system

Spring–Mass–Damper System Parameters

The standard transfer function of a mass–spring–damper system (with force input $F(s)$ and displacement output $X(s)$) is given by:

$$G_{ms}(s) = = \frac{1}{ms^2 + cs + k}.$$

Dividing by m:

$$G_{ms}(s) = \frac{\frac{1}{m}}{s^2 + \frac{c}{m}s + \frac{k}{m}}.$$

Comparison with Standard Second-Order Form

The general second-order transfer function is written as:

$$G(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

By comparing denominators:

$$\frac{c}{m} = 2\zeta\omega_n, \qquad \frac{k}{m} = \omega_n^2.$$

Parameter Values (with $m = 1$)

Given:

$$\zeta = 0.49999282386, \qquad \omega_n = 1.450970082, \qquad m = 1,$$

we obtain:

$$k = \omega_n^2 = (1.450970082)^2 \approx 2.105314179,$$

$$c = 2\zeta\omega_n = 2(0.49999282386)(1.450970082) \approx 1.450949257.$$

Final Parameters

$$m = 1, \qquad c \approx 1.4509, \qquad k \approx 2.1053.$$

Thus, the equivalent spring–mass–damper system has stiffness $k \approx 2.1053$ and damping coefficient $c \approx 1.4509$ for a unit mass.

Damping Coefficient	Spring coefficient
1.4509	2.1053

Table 2: System Identification Values from the calculations

Which has the following Bode Plot

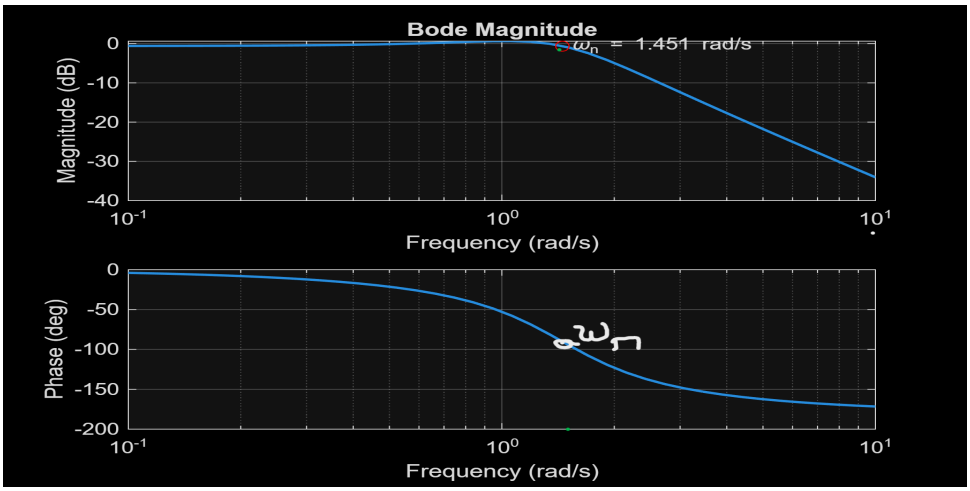

Figure 4: Bode plot of the System derived from the step response

In Figure 4, the magnitude curve shows how the system output amplitude and phase lag differ with changes in frequency. From these important parameters, i.e, corner frequency, resonant peak, and gain margin, can be determined, offering insight into the system stability and dynamic behaviour.

`Now, Poles calculations:`

Poles of a system play a crucial role in understanding its stability and dynamic response. They are found as follows:

The poles are the roots of the denominator,

$$s^2 + 1.44s + 2.10 = 0$$

Using the quadratic formula:

$$s = \frac{-1.44 \pm \sqrt{1.44^2 - 4(1)(2.10)}}{2}$$

$$s = \frac{-1.44 \pm \sqrt{-6.3264}}{2}$$

$$s = \frac{-1.44}{2} \pm j\frac{2.515}{2}$$

$$s = -0.725 \pm j1.257$$

which can be drawn in the s-plane as follows:

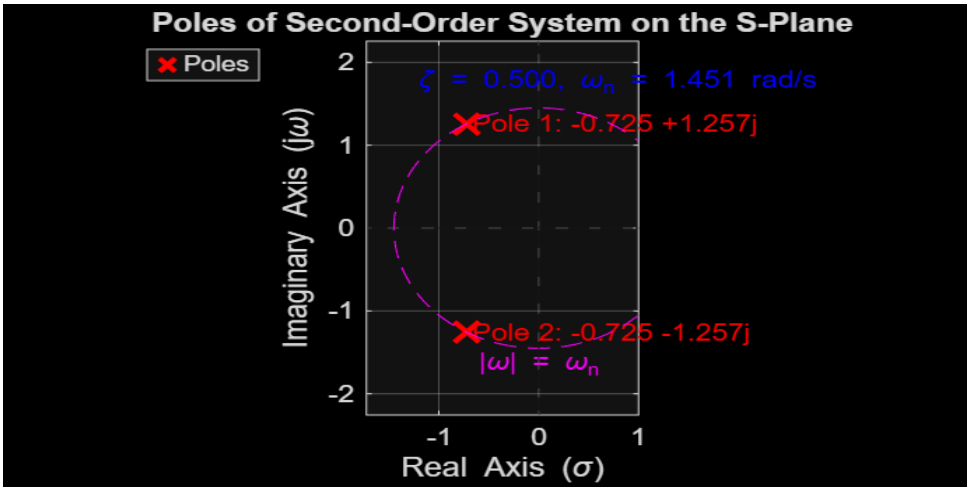

Figure 5: poles of the derived system equation

Figure 5 shows that this system has a natural frequency and on the left side of the s-plane, therefore concluding that the system is underdamped and stable.

3.2 Frequency Response Stage 2

In this stage, the suspension system is characterized by analysing its frequency response. Sinusoidal inputs of varying frequencies are applied, and the resulting outputs are examined to determine the system's behaviour. The response from Stage 1 is verified by comparing the amplitude and phase shifts between the input and output signals, and a preliminary Bode plot is constructed to highlight the key frequency-domain characteristics.

Figure 6 shows the Bode plot generated using the collected frequencies from 0.1 rad/s to 4 rad/s. This frequency range is sufficient to describe the system's behavior. This helps us to identify important details about the system, i.e, resonant frequency and gain and phase(lead or lag) at various frequencies This plot further confirms the dominant poles of the system and its frequency response characteristic.

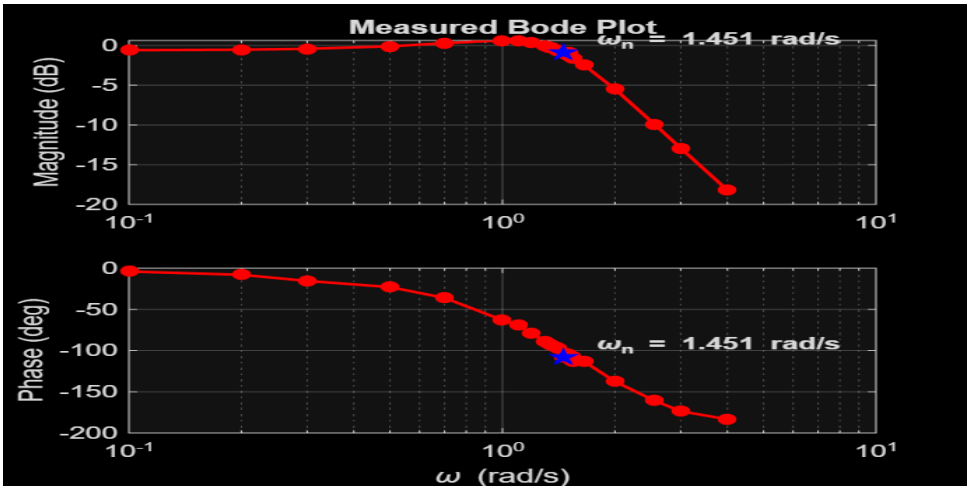

Figure 6: Bode plot of the System derived from the experiment collected data

3.3 Model Validation

In this section, the validation of the graphs is performed by comparing the system's simulated step and frequency responses with measurements taken during laboratory experiments. So the target is to determine how closely the simulation matches the real-world performance, which gives us insight into the model's accuracy, together with its usefulness in representing the system.

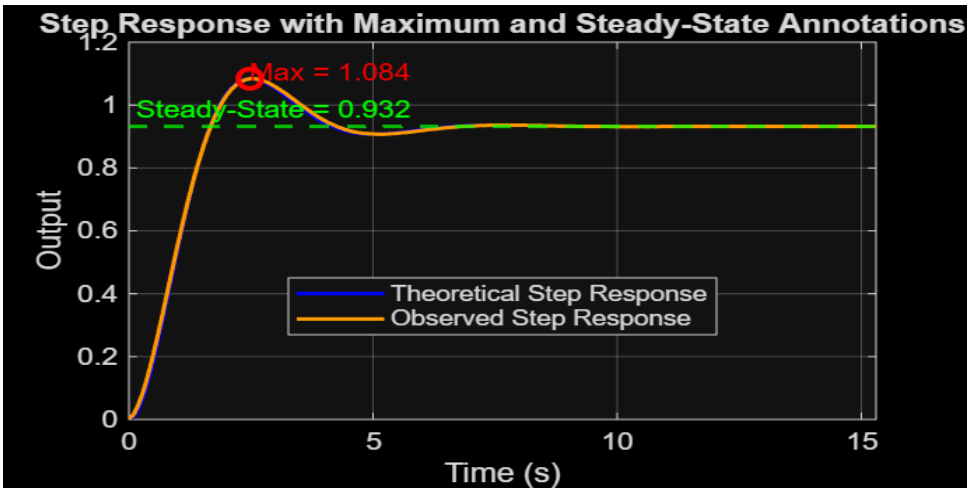

Figure 7: Derived Response and the Lab Response

Figure 7 above shows the two-step responses in one plane, one is from the Derived response generated through the use of a transfer function, and the other is the Lab Response, which was obtained from the experimental data collected during lab testing. Both curves share the same peak value, with very similar response times. Importantly, they both converge to the same steady-state gain of 0.932, confirming the accuracy of the derived model..

```
Now, Bode plot validation
```

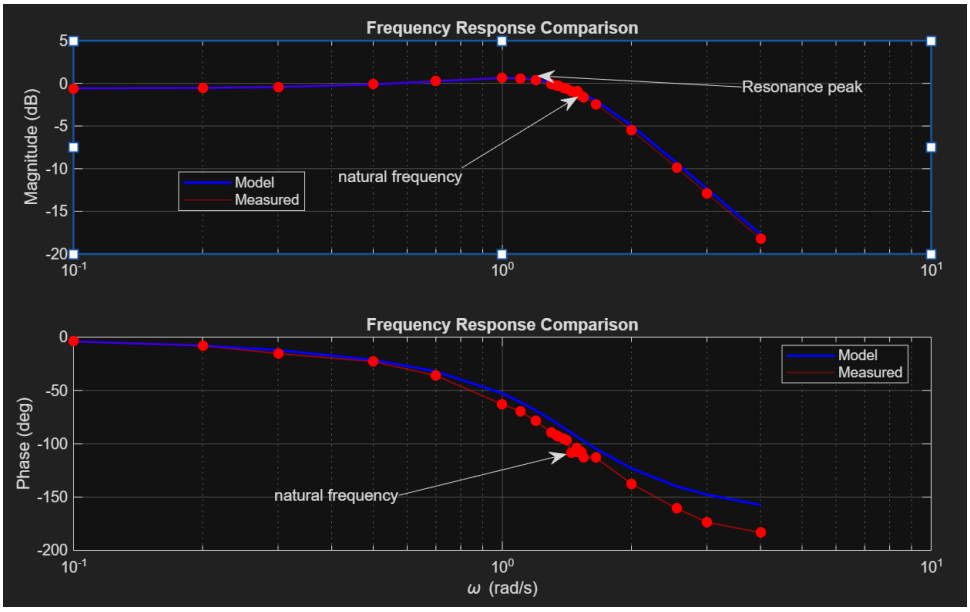

Figure 8: Bode plot of Derived Response and the Lab Response comparison

In Figure 8, both the Bode plots of the Measured and Model transfer functions are compared. From this figure, the Bode magnitude plot obtained from the measured data is consistent with the plot calculated from the transfer function model in the system identification equation. Figure 8 also shows Bode phase plots of the derived and experimental transfer functions. The one made from red dots connected by a faint red line corresponding to the phase of the experimental data aligns closely with the derived data, although there are differences between them here and there, which is due to inaccuracies, especially the more as frequency increases, the more inaccurate the alignment becomes.

`Validation and limitations:`

1. Frequency Response Validation: Experimental and model Bode magnitude plots showed strong agreement. Phase discrepancies at higher frequencies were due to simulator simplifications and modelling assumptions.
2. Step Response Validation: Experimental and simulated responses matched closely in peak, response time, and steady-state gain.

`Improvements to counteract limitations :`

Model Refinement: Adjusting the natural frequency to 1.1 rad/s improved alignment between simulated and experimental phase plots, reducing lag differences.

4 Conclusion

The second-order system model identified captures the dominant dynamics of the suspension: with a damping ratio $\zeta \approx 0.50$ and natural frequency $\omega_n \approx 1.45$ rad/s . This model describes the frequency behaviour and the transient, and it is good for control design together with preliminary analysis. Model limitations at higher frequencies are noted and discussed.

5 References

1. University of Cape Town, EEE3094S Lab 1 User Manual, Amathuba, 2023. [Online]. Available: https://amathuba.uct.ac.za . [Accessed: 03-Sep-2023].
2. @onlinewikipedia2023, author = Wikipedia contributors, title = Mass-spring-damper model, year = 2023, url = https://en.wikipedia.org/wiki/Mass-spring-damper$_{m}odel, note = [Accessed 3 Sep.$2023]

A Appendix and Notes

```
%% ------------------- Transfer Function ----------------------
% Define system parameters for a standard second-order system
A    = 0.932;                % System gain
Zeta = 0.49999282386;        % Damping ratio (ζ)
wn   = 1.450970082;          % Natural frequency (rad/s)

% Compile transfer function
wnSq   = wn^2;               % Square of natural frequency
num    = A * wn^2;           % Numerator of TF (gain*wn^2)
bCoef  = 2 * Zeta * wn;      % Damping coefficient in denominator
den    = [1 bCoef wnSq];     % Denominator of TF (s^2 + 2ζω_ns + ω_n^2)
G      = tf(num, den);       % Create transfer function object

%% ------------------- Step Response ----------------------
% Create time vector for simulation
t_end = 10;                  % End time (s)
dt    = 0.01;                % Time step (s)
time  = 0:dt:t_end;          % Time vector for step response plot

% Plot step response of the system
figure;
step(G, time);
title('Step Response of Model');
xlabel('Time (s)');
ylabel('Output');
grid on;

%% ------------------- Frequency Response ----------------------
% Load measured frequency response from Excel
FR = readtable("Frequency response.xlsx");

% Use MATLAB-friendly column names (automatically cleaned)
f_exp_rad     = FR.f_rads_sec_;     % Frequency points (rad/s)
mag_exp_dB    = FR.Gain_dB_;        % Measured magnitude (dB)
phase_exp_deg = FR.Phase_degrees_;  % Measured phase (degrees)

% Compute model response at the measured frequencies
[mag_model, phase_model] = bode(G, f_exp_rad); % Bode output in magnitude & phase
mag_model   = squeeze(mag_model);              % Remove singleton dimensions
phase_model = squeeze(phase_model);           % Remove singleton dimensions
mag_model_dB = 20*log10(mag_model);           % Convert magnitude to dB

% Plot Bode comparison between model and measured data
```

```
figure;

% --- Magnitude plot ---
subplot(2,1,1);
semilogx(f_exp_rad, mag_model_dB, 'b','LineWidth',1.5); hold on; % Model in blue
semilogx(f_exp_rad, mag_exp_dB, '-','Color',[1 0 0 0.3],'LineWidth',1); % Measured fa
semilogx(f_exp_rad, mag_exp_dB, 'or','MarkerFaceColor','r');  % Measured points in re
ylabel('Magnitude (dB)');
grid on;
legend('Model','Measured','Location','best');
title('Frequency Response Comparison');

% --- Phase plot ---
subplot(2,1,2);
semilogx(f_exp_rad, phase_model, 'b','LineWidth',1.5); hold on;  % Model in blue
semilogx(f_exp_rad, phase_exp_deg, '-','Color',[1 0 0 0.3],'LineWidth',1); % Measured
semilogx(f_exp_rad, phase_exp_deg, 'or','MarkerFaceColor','r'); % Measured points in
xlabel('\omega (rad/s)');
ylabel('Phase (deg)');
grid on;
legend('Model','Measured','Location','best');
% ENTER THE FOLLOWING
A = 0.932;
Zeta = 0.49999282386;
wn = 1.450970082;

% COMPILE TRANSFER FUNCTION
wnSq = wn^2;
num = A * wn^2;
bCoef = 2 * Zeta * wn;
den = [1 bCoef wnSq];
G = tf(num, den);

% STEP SIMULATION
step(G);
%bode(G);

% STEP REAL/OBSERVED RESPONSE
hold on
plot(F34BSuspensionTestData,"Time","Output","Color",[1 0.6 0]);

% FIND ROOTS
r = roots(den);
% Step Input Signal with Full Annotations
% Author: Mbhekwa Ntuli - EEE3094S Lab 1
```

```
% --- Time and Input ---
t = -1:0.01:10;          % time vector (starts before zero to show baseline)
u = double(t >= 0);      % unit step: 0 for t<0, 1 for t>=0

% --- Create Figure ---
fig = figure('Units','inches','Position',[1 1 7 3.5]);
plot(t, u, 'b-', 'LineWidth', 2); hold on;
grid on;

% --- Labels ---
xlabel('Time (s)', 'FontSize', 11);
ylabel('Input Amplitude', 'FontSize', 11);
title('Unit Step Input Signal', 'FontSize', 12);
xlim([min(t) max(t)]);
ylim([-0.2 1.2]);
set(gca, 'FontSize', 10);

% --- Annotations ---
% Mark the step at t=0
plot(0, 1, 'ro', 'MarkerFaceColor','r', 'MarkerSize', 7);
text(0.2, 0.6, 'Step applied at t = 0 s', 'Color','r','FontSize',10);

% Steady-state line
plot([min(t) max(t)], [1 1], '--k', 'LineWidth', 1.2);
text(1, 1.05, 'A = 1 (steady-state)', 'FontSize', 10, 'Color','k');

% Arrow annotation pointing to the step
annotation('textarrow',[0.35 0.2],[0.5 0.8],...
    'String','Discontinuity at t=0','FontSize',10);

% --- Export the figure as PNG (300 dpi) ---
print(fig,'step_input_annotated.png','-dpng','-r300');

disp('Figure saved as step_input_annotated.png');
```

A.1 MATLAB Code

```
%% ------------------- Transfer Function ----------------------
% Define system parameters for a standard second-order system
A    = 0.932;                % System gain
Zeta = 0.49999282386;        % Damping ratio (ζ)
wn   = 1.450970082;          % Natural frequency (rad/s)

% Compile transfer function
```

```
wnSq   = wn^2;                % Square of natural frequency
num    = A * wn^2;            % Numerator of TF (gain*wn^2)
bCoef  = 2 * Zeta * wn;       % Damping coefficient in denominator
den    = [1 bCoef wnSq];      % Denominator of TF (s^2 + 2ζω_ns + ω_n^2)
G      = tf(num, den);        % Create transfer function object

%% ------------------- Step Response ----------------------
% Create time vector for simulation
t_end = 10;                   % End time (s)
dt    = 0.01;                 % Time step (s)
time  = 0:dt:t_end;           % Time vector for step response plot

% Plot step response of the system
figure;
step(G, time);
title('Step Response of Model');
xlabel('Time (s)');
ylabel('Output');
grid on;

%% ------------------- Frequency Response ----------------------
% Load measured frequency response from Excel
FR = readtable("Frequency response.xlsx");

% Use MATLAB-friendly column names (automatically cleaned)
f_exp_rad     = FR.f_rads_sec_;     % Frequency points (rad/s)
mag_exp_dB    = FR.Gain_dB_;        % Measured magnitude (dB)
phase_exp_deg = FR.Phase_degrees_;  % Measured phase (degrees)

% Compute model response at the measured frequencies
[mag_model, phase_model] = bode(G, f_exp_rad); % Bode output in magnitude & phase
mag_model   = squeeze(mag_model);              % Remove singleton dimensions
phase_model = squeeze(phase_model);           % Remove singleton dimensions
mag_model_dB = 20*log10(mag_model);           % Convert magnitude to dB

% Plot Bode comparison between model and measured data

figure;

% --- Magnitude plot ---
subplot(2,1,1);
semilogx(f_exp_rad, mag_model_dB, 'b','LineWidth',1.5); hold on;
% Model response in blue line
% This is the theoretical magnitude (in dB) calculated from the transfer function
semilogx(f_exp_rad, mag_exp_dB, '-','Color',[1 0 0 0.3],'LineWidth',1);
```

```
% Faint red line connecting measured points
% Helps visualize the trend without overpowering the model curve
semilogx(f_exp_rad, mag_exp_dB, 'or','MarkerFaceColor','r');
% Measured data points in red with filled circles
ylabel('Magnitude (dB)');
grid on;
legend('Model','Measured','Location','best');
title('Frequency Response Comparison');

% --- Phase plot ---
subplot(2,1,2);
semilogx(f_exp_rad, phase_model, 'b','LineWidth',1.5); hold on;
% Model phase in blue line
semilogx(f_exp_rad, phase_exp_deg, '-','Color',[1 0 0 0.3],'LineWidth',1);
% Faint red line connecting measured phase points
semilogx(f_exp_rad, phase_exp_deg, 'or','MarkerFaceColor','r');
% Measured phase points in red with filled circles
xlabel('\omega (rad/s)');
ylabel('Phase (deg)');
grid on;
legend('Model','Measured','Location','best');
title('Frequency Response Comparison');

% --- Phase plot ---
subplot(2,1,2);
semilogx(f_exp_rad, phase_model, 'b','LineWidth',1.5); hold on;  % Model in blue
semilogx(f_exp_rad, phase_exp_deg, '-','Color',[1 0 0 0.3],'LineWidth',1); % Measured
semilogx(f_exp_rad, phase_exp_deg, 'or','MarkerFaceColor','r'); % Measured points in
xlabel('\omega (rad/s)');
ylabel('Phase (deg)');
grid on;
legend('Model','Measured','Location','best');
```